

Node.js Developer – Sample Course Syllabus

Module 1: Introduction to Node.js

- What is Node.js
- Node.js Architecture (Event-driven, Non-blocking I/O)
- Installing Node.js and npm
- Running the first Node.js application

Module 2: Node.js Fundamentals

- Node REPL
- Modules and require()
- Creating custom modules
- Global objects and utilities

Module 3: Built-in Modules

- File System (fs) module
- HTTP module
- Path module
- OS module

Module 4: Web Servers with Node.js

- Creating a basic HTTP server
- Handling requests and responses
- Serving HTML and JSON

Module 5: Express.js Framework

- Introduction to Express
- Routing (GET, POST, PUT, DELETE)
- Middleware concept
- Building REST APIs

Module 6: Database Integration

- Connecting Node.js with MongoDB
- Using Mongoose ORM

- CRUD operations

Module 7: Authentication & Security

- User login and registration
- Password hashing with bcrypt
- Authentication using JWT

Module 8: File Upload & Advanced Concepts

- File upload using Multer
- Error handling
- Environment variables
- API testing with Postman

Module 9: Deployment

- Preparing Node.js apps for production
- Deploying on cloud platforms
- Basic DevOps concepts

Final Projects

- Student Management API
- Blog REST API
- Authentication system
- Full stack project (React + Node.js)